9 A company orders a number of bottles from a factory.

The 8 machines in the factory could make all the bottles in 5 days.
All the machines work at the same rate.

For 2 days, only 4 machines are used to make the bottles.
From the 3rd day, all 8 machines are used to make the bottles.

Work out the total number of days taken to make all the bottles.

Total work = 8 × 5 = 40 machine-days.

First 2 days: 4 × 2 = 8 machine-days completed.

1 mark
Find remaining work

Work remaining = 40 − 8 = 32 machine-days.

At 8 machines: 32 ÷ 8 = 4 more days.

1 mark
Finish at 8 machines then add first 2 days

Total time = 2 + 4 = 6 days.

1 mark
6 days

..... days

(Total for Question 9 is 3 marks)